Scientific Context of the Mexican Hat Scalar Potential Framework

🌌 Scientific Context

🎯 Mexican Hat Potential

Represents a spontaneous symmetry-breaking scalar field potential:

$$V(\phi) = \lambda(\phi^2 - v^2)^2$$

Perturbations from GW events are injected as:

$$V(z) = V_0(\phi(z)) + \delta \cdot J\_injection(z)$$

The second derivative V″(z) modulates acceleration:

$$\ddot{a}(z)/a(z) \sim -V''(z)$$

🌀 Kerr–de Sitter Black Hole Framework

- The Universe lies inside a rotating black hole.
- Mass accretion events change its angular momentum and curvature structure.
- These changes deform the scalar potential saddle, mimicking dark energy.

✅ Conclusion

This program demonstrates that:

- Black hole mergers, as observed via GWs, are not just astrophysical curiosities.
- They may be shaping the large-scale expansion of the Universe by modulating scalar potential curvature.
- Offering a viable physical explanation for mid-redshift acceleration — without Λ.